

Regulation of Mitochondrial ATP Production: Ca²⁺ Signaling and Quality Control.

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Cardiac ATP production primarily depends on oxidative phosphorylation in mitochondria and is dynamically regulated by Ca²⁺ levels in the mitochondrial matrix as well as by cytosolic ADP. We discuss mitochondrial Ca²⁺ signaling and its dysfunction which has recently been linked to cardiac pathologies including arrhythmia and heart failure. Similar dysfunction in other excitable and long-lived cells including neurons is associated with neurodegenerative diseases such as Alzheimer's disease (AD), amyotrophic lateral sclerosis (ALS), and Parkinson's disease (PD). Central to this new understanding is crucial Ca²⁺ regulation of both mitochondrial quality control and ATP production. Mitochondria-associated membrane (MAM) signaling from the sarcoplasmic reticulum (SR) and the endoplasmic reticulum (ER) to mitochondria is discussed. We propose future research directions that emphasize a need to define quantitatively the physiological roles of MAMs, as well as mitochondrial quality control and ATP production.

Calcium Overload and Mitochondrial Metabolism.

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Mitochondria calcium is a double-edged sword. While low levels of calcium are essential to maintain optimal rates of ATP production, extreme levels of calcium overcoming the mitochondrial calcium retention capacity leads to loss of mitochondrial function. In moderate amounts, however, ATP synthesis rates are inhibited in a calcium-titratable manner. While the consequences of extreme calcium overload are well-known, the effects on mitochondrial function in the moderately loaded range remain enigmatic. These observations are associated with changes in the mitochondria ultrastructure and cristae network. The present mini review/perspective follows up on previous studies using well-established cryo-electron microscopy and poses an explanation for the observable depressed ATP synthesis rates in mitochondria during calcium-overloaded states. The results presented herein suggest that the inhibition of oxidative phosphorylation is not caused by a direct decoupling of energy metabolism via the opening of a calcium-sensitive, proteinaceous pore but rather a separate but related calcium-dependent phenomenon. Such inhibition during calcium-overloaded states points towards mitochondrial ultrastructural modifications, enzyme activity changes, or an interplay between both events.

ROS production by mitochondria: function or dysfunction?

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In eukaryotic cells, ATP generation is generally viewed as the primary function of mitochondria under normoxic conditions. Reactive oxygen species (ROS), in contrast, are regarded as the by-products of respiration, and are widely associated with dysfunction and disease. Important signaling functions have been demonstrated for mitochondrial ROS in recent years. Still, their chemical reactivity and capacity to elicit oxidative damage have reinforced the idea that ROS are the products of dysfunctional mitochondria that accumulate during disease. Several studies support a different model, however, by showing that: (1) limited oxygen availability results in mitochondria prioritizing ROS production over ATP, (2) ROS is an essential adaptive mitochondrial signal triggered by various important stressors, and (3) while

mitochondria-independent ATP production can be easily engaged by most cells, there is no known replacement for ROS-driven redox signaling. Based on these observations and other evidence reviewed here, we highlight the role of ROS production as a major mitochondrial function involved in cellular adaptation and stress resistance. As such, we propose a rekindled view of ROS production as a primary mitochondrial function as essential to life as ATP production itself.

Cellular ATP demand creates metabolically distinct subpopulations of mitochondria.

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Mitochondria serve a crucial role in cell growth and proliferation by supporting both ATP synthesis and the production of macromolecular precursors. Whereas oxidative phosphorylation (OXPHOS) depends mainly on the oxidation of intermediates from the tricarboxylic acid cycle, the mitochondrial production of proline and ornithine relies on reductive synthesis¹. How these competing metabolic pathways take place in the same organelle is not clear. Here we show that when cellular dependence on OXPHOS increases, pyrroline-5-carboxylate synthase (P5CS)-the rate-limiting enzyme in the reductive synthesis of proline and ornithine-becomes sequestered in a subset of mitochondria that lack cristae and ATP synthase. This sequestration is driven by both the intrinsic ability of P5CS to form filaments and the mitochondrial fusion and fission cycle. Disruption of mitochondrial dynamics, by impeding mitofusin-mediated fusion or dynamin-like-protein-1-mediated fission, impairs the separation of P5CS-containing mitochondria from mitochondria that are enriched in cristae and ATP synthase. Failure to segregate these metabolic pathways through mitochondrial fusion and fission results in cells either sacrificing the capacity for OXPHOS while sustaining the reductive synthesis of proline, or foregoing proline synthesis while preserving adaptive OXPHOS. These findings provide evidence of the key role of mitochondrial fission and fusion in maintaining both oxidative and reductive biosyntheses in response to changing nutrient availability and bioenergetic demand.

Mitochondria: roles in pulmonary hypertension.

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Mitochondria are essential cell organelles responsible for ATP production in the presence of oxygen. In the pulmonary vasculature, mitochondria contribute to physiological intracellular signalling pathways through production of reactive oxygen species and play the role of oxygen sensors that coordinate hypoxic pulmonary vasoconstriction. Mitochondria also play a pathophysiological role in pulmonary hypertension (PH). This disease is characterized by increased pulmonary arterial pressure and remodelling of pulmonary arteries, leading to increased pulmonary vascular resistance, hypertrophy of the right ventricle, right heart failure and ultimately death. Mitochondrial alterations have been evidenced in PH in pulmonary arteries and in the right ventricle, in particular a chronic shift in energy production from mitochondrial oxidative phosphorylation to glycolysis. This shift, initially described in cancer cells, may play a central role in PH pathogenesis. Further studies of these metabolic mitochondrial alterations in PH may therefore open new therapeutic perspectives in this disease.

Acute Imidacloprid Exposure Alters Mitochondrial Function in Bumblebee Flight Muscle and Brain.

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Mitochondria are intracellular organelles responsible for cellular respiration with one of their major roles in the production of energy in the form of ATP. Activities with increased energetic demand are especially dependent on efficient ATP production, hence sufficient mitochondrial function is fundamental. In bees, flight muscle and the brain have particularly high densities of mitochondria to facilitate the substantial ATP production required for flight activity and neuronal signalling. Neonicotinoids are systemic synthetic insecticides that are widely utilised against crop herbivores but have been reported to cause, by unknown mechanisms, mitochondrial dysfunction, decreasing cognitive function and flight activity among pollinating bees. Here we explore, using high-resolution respirometry, how the neonicotinoid imidacloprid may affect oxidative phosphorylation in the brain and flight muscle of the buff-tailed bumblebee, *Bombus terrestris*. We find that acute exposure increases routine oxygen consumption in the flight muscle of worker bees. This provides a candidate explanation for prior reports of early declines in flight activity following acute exposure. We further find that imidacloprid increases the maximum electron transport capacity in the brain, with a trend towards increased overall oxygen consumption. However, intra-individual variability is high, limiting the extent to which apparent effects of imidacloprid on brain mitochondria are shown conclusively. Overall, our results highlight the necessity to examine tissue-specific effects of imidacloprid on respiration and energy production.

Mitochondrial connections with immune system in Zebrafish.

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Mitochondria are organelles commonly associated with adenosine triphosphate (ATP) formation through the oxidative phosphorylation (OXPHOS) process. However, mitochondria are also responsible for functions such as calcium homeostasis, apoptosis, autophagy, and production of reactive oxygen species (ROS) that, in conjunction, can lead to different cell fate decisions. Mitochondrial morphology changes rely on nutrients' availability and the bioenergetics demands of the cells, in a process known as mitochondrial dynamics, which includes both fusion and fission. This organelle senses the microenvironment and can modify the cells to either a pro or anti-inflammatory profile. The zebrafish has been increasingly used to research mitochondrial dynamics and its connection with the immune system since the pathways and molecules involved in these processes are conserved on this fish. Several genetic tools and technologies are currently available to analyze the behavior of mitochondria in zebrafish. However, even though zebrafish presents several similar processes known in mammals, the effect of the mitochondria in the immune system has not been so broadly studied in this model. In this review, we summarize the current knowledge in zebrafish studies regarding mitochondrial function and immuno metabolism.

Somatic mutations of mitochondrial DNA in aging and cancer progression.

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Mitochondria are intracellular organelles responsible for generating ATP through respiration and oxidative

phosphorylation (OXPHOS), producing reactive oxygen species, and initiating and executing apoptosis. Mitochondrial dysfunction has been observed to be an important hallmark of aging and cancer. Because mitochondrial DNA (mtDNA) is important in maintaining functionally competent organelles, accumulation of mtDNA mutations can affect energy production, oxidative stress, and cell survival, which may contribute to aging and/or carcinogenesis. This review outlines a variety of somatic mtDNA mutations identified in aging tissues and human cancers, as well as recent advances in understanding the causal role of mtDNA mutations in the aging process and cancer progression. Mitochondrial dysfunction elicited by somatic mutations in mtDNA could induce apoptosis in aging cells and some cancer cells with severe mtDNA mutations. In addition, it could activate mitochondria-to-nucleus retrograde signaling to modulate the expression of nuclear genes involved in a metabolic shift from OXPHOS to glycolysis, facilitate cells to adapt to altered environments and develop resistance to chemotherapeutic agents, or promote metastatic properties of cancer cells. These findings suggest that accumulation of somatic mtDNA mutations is not only an important contributor to human aging but also plays a critical role in cancer progression.

Assessing mitochondrial respiratory bioenergetics in whole cells and isolated organelles by microplate respirometry.

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Mitochondria are responsible for the generation of ATP by oxidative phosphorylation; however, these multifaceted organelles regulate many other key cellular functions as well, such as calcium homeostasis, apoptosis, and biosynthesis of steroid hormones, heme and phospholipids. In order to carry out these functions, mitochondria establish physical and functional connections with other organelles such as the plasma membrane, lipid droplets/vesicles, peroxisomes, endosomes, and the endoplasmic reticulum. Dysregulation of any of the aforementioned processes or inter-organelle contacts can lead to mitochondrial dysfunction and subsequent changes in oxygen consumption and ATP production. Seahorse technology has become a critical tool for quantification of mitochondrial oxygen consumption and can help differentiate primary mitochondrial disorders from disorders where alterations in mitochondrial metabolism are consequences of a prior, upstream insult. In this chapter, we describe the application of Seahorse technology for assaying mitochondrial respiration in whole cells, permeabilized cells and isolated mitochondria. We leave it to the researcher's discretion to determine which of these approaches will generate the most physiologically relevant data based on the model system and research question at hand.

Mitochondrial Metabolism in Major Neurological Diseases.

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Mitochondria are bilayer sub-cellular organelles that are an integral part of normal cellular physiology. They are responsible for producing the majority of a cell's ATP, thus supplying energy for a variety of key cellular processes, especially in the brain. Although energy production is a key aspect of mitochondrial metabolism, its role extends far beyond energy production to cell signaling and epigenetic regulation—functions that contribute to cellular proliferation, differentiation, apoptosis, migration, and autophagy. Recent research on neurological disorders suggest a major metabolic component in disease

pathophysiology, and mitochondria have been shown to be in the center of metabolic dysregulation and possibly disease manifestation. This review will discuss the basic functions of mitochondria and how alterations in mitochondrial activity lead to neurological disease progression.
